

PROLAB – KF, EKF, PF

Evaluation unter varying process and measurement noise

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Introduction

- Implementation of
 - Kalman Filter (KF)
 - Extended Kalman Filter (EKF)
 - EKF with landmark detection (EKF-LM)
 - Particle Filter (PF)
- Specific Task: “Systematically analyze the effect of Q vs R tuning.”
- Setup:
 - ROS2 Foxy
 - TurtleBot3 Waffle
 - Gazebo Classic

Methods – Noisy Odometry

- */odom* is used as the ground truth
- Real-world conditions (noisy encoders) are simulated
- The *joint_states* are extracted and Gaussian Noise is added
- A noisy odometry is calculated with this data
 - *noisy_odometry_node*

Methods – Kalman Filter

- *Concept*

- Strictly linear dynamics
- Decoupled position-velocity state (x, y, v_x, v_y)
- Completely omits orientation θ

- *Data Flow & Inputs*

- **Prediction.** No active control input vector; State propagation relies on state transition (velocity from the previous state)
- **Correction.** Position updates from *noisy_odometry*

Methods – Extended Kalman Filter

- *Concept*

- Non-linear state estimator
- First-order Taylor series expansion (Jacobian matrices)
- Tracks full planar pose x, y, θ

- *Data Flow & Inputs*

- **Prediction.** *noisy_odometry used as velocity input*
- **Correction.** IMU updates yaw/heading data

Methods – Landmark Detection

- *Concept*

- Custom Gazebo World with 3 distinct cylinders
- LIDAR-Scans are used
- Groups of 3 are picked of the laserscans
- Circle diameters are compared
- Voting system, tolerances

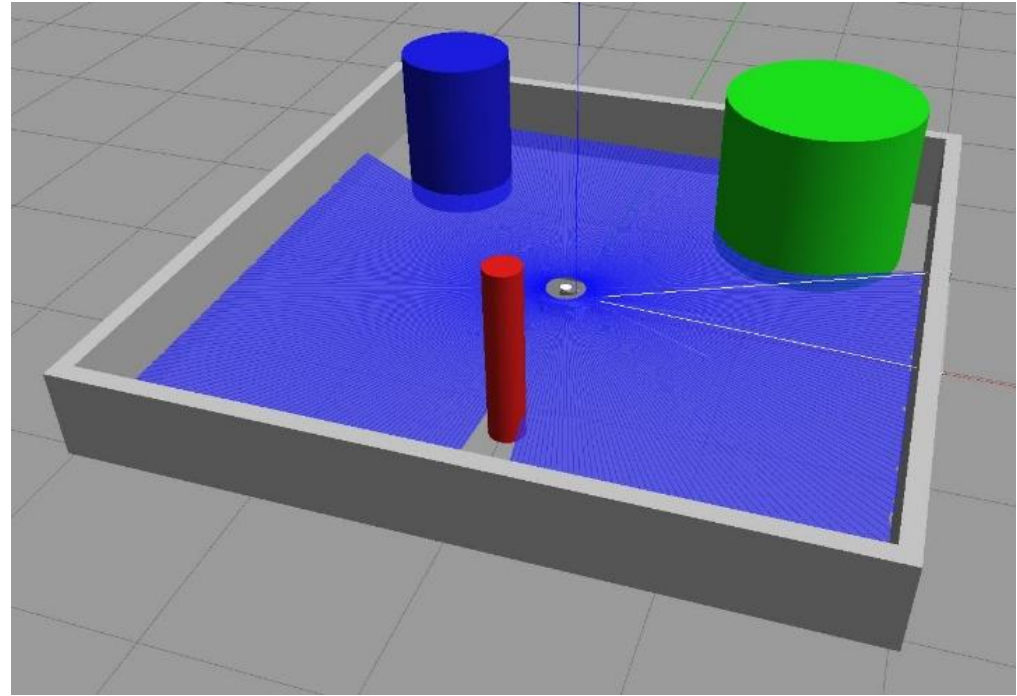


Fig. 1: The designed Gazebo world with three distinct cylinders, where the blue one serves as a landmark.

Methods – Landmark Detection

- *Limitations*

- The biggest cylinder ($r = 600\text{mm}$) is hard to distinguish from the wall
- Evaluating LIDAR-Scans is computationally expensive
- Parameters are introduced to reduce cost: *scan_skip*, *stride*
- Smallest cylinder ($r = 100\text{mm}$) might be not noticed

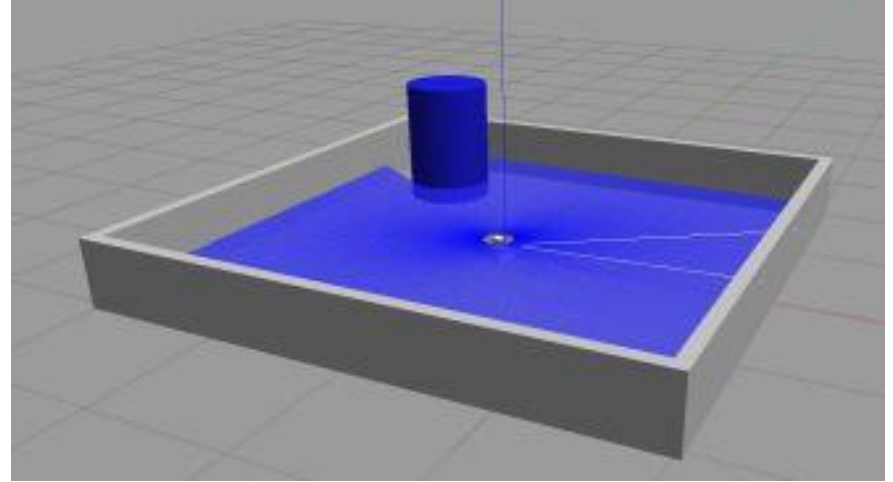


Fig. 2: The blue cylinder serves as a landmark.



Only one cylinder serves as a landmark ($r = 350\text{mm}$)

Methods – Landmark Detection

- *Geometric limitations?*
 - When the robot is standing still, its pose is not unique with only one landmark used
 - Infinite positions with the same angle between robot and rotationally symmetric cylinder
- *No geometric limitations!*
 - However, the robot is not standing still
 - The landmark detection is integrated in the EKF

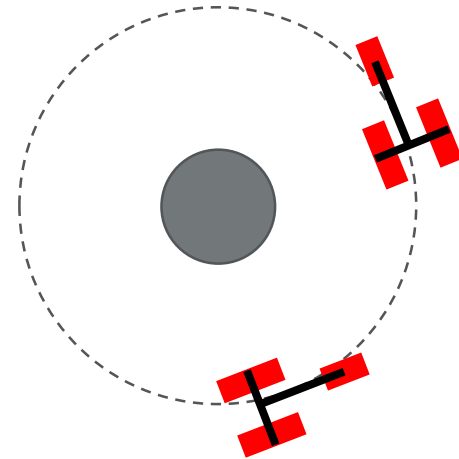


Fig. 3: A rotationally symmetric cylinder looks the same to the robot at infinite poses on a circle.

Methods – Particle Filter

- *Concept*

- Sample-based state estimator
- Swarm of 1,500 hypotheses
- Capable of representing non-Gaussian, multimodal probability distributions

- *Data Flow & Inputs*

- **Prediction.** *Direct forward control commands /cmd_vel*
- **Correction.** Laser scan likelihood fields → object detection for plausibility check

Methods – Experimental setup, Evaluation

- Identical trajectories within the custom Gazebo world
- 5 zero-radius-turns
- Trajectories are created by directly controlled velocity commands
 - Same input throughout the trials
- Evaluation criteria:
 - Root Mean Squared Error (RMSE) between estimated trajectories and ground truth
 - Euclidean distance between endpoints of estimated trajectory and ground truth

Methods – Experimental setup, Evaluation

- Base values for noise of filters are defined, e.g.:
 - $\text{base_r_imu} = 0.05 \text{ rad}$ (standard deviation of θ ; equals 2.86°)
- Factor_Q and Factor_R are multiplied to every noise value
- Order-of-magnitude scaling factors
 - 0.01 low
 - 1.0 standard
 - 10.0 intermediate
 - 100.0 extreme



Four evaluation groups

Results – Example of plotted trajectories

Comparison of the estimators

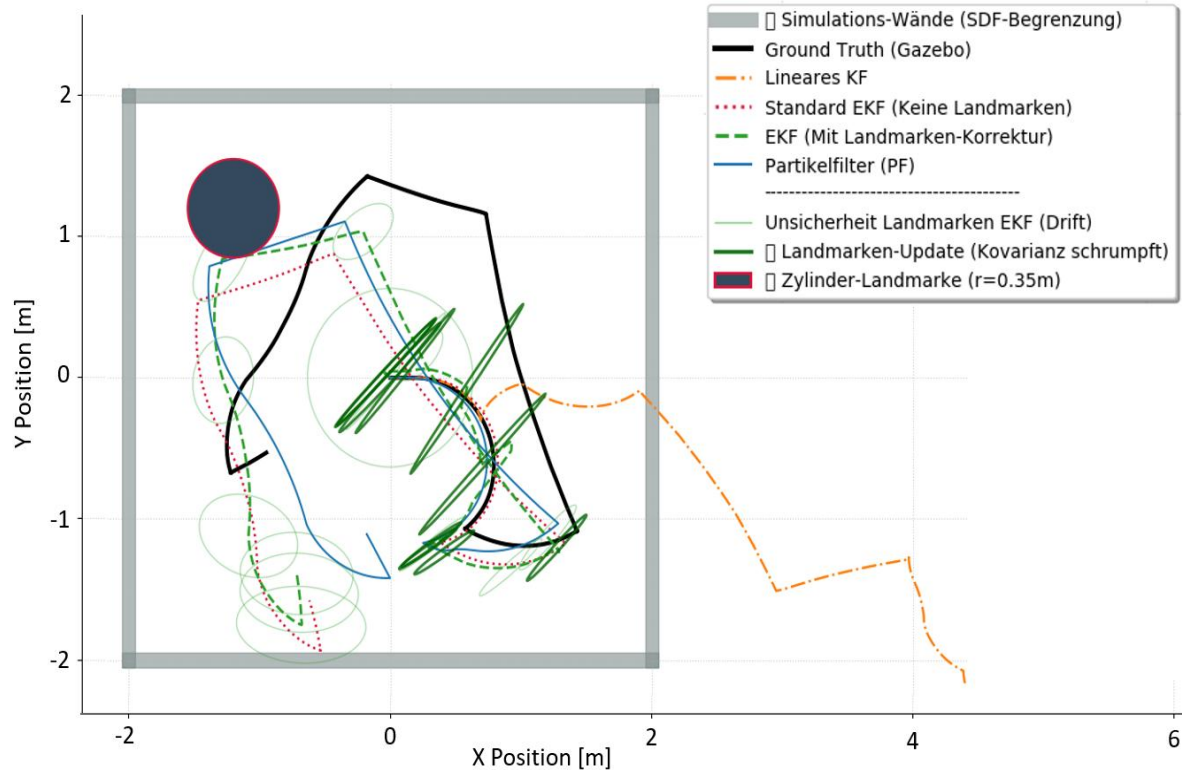


Fig. 4: Plotted trajectories of the experimental run with $Q = R = 0.01$.

Results – Evaluation Group 1: Minimal Measurement Noise

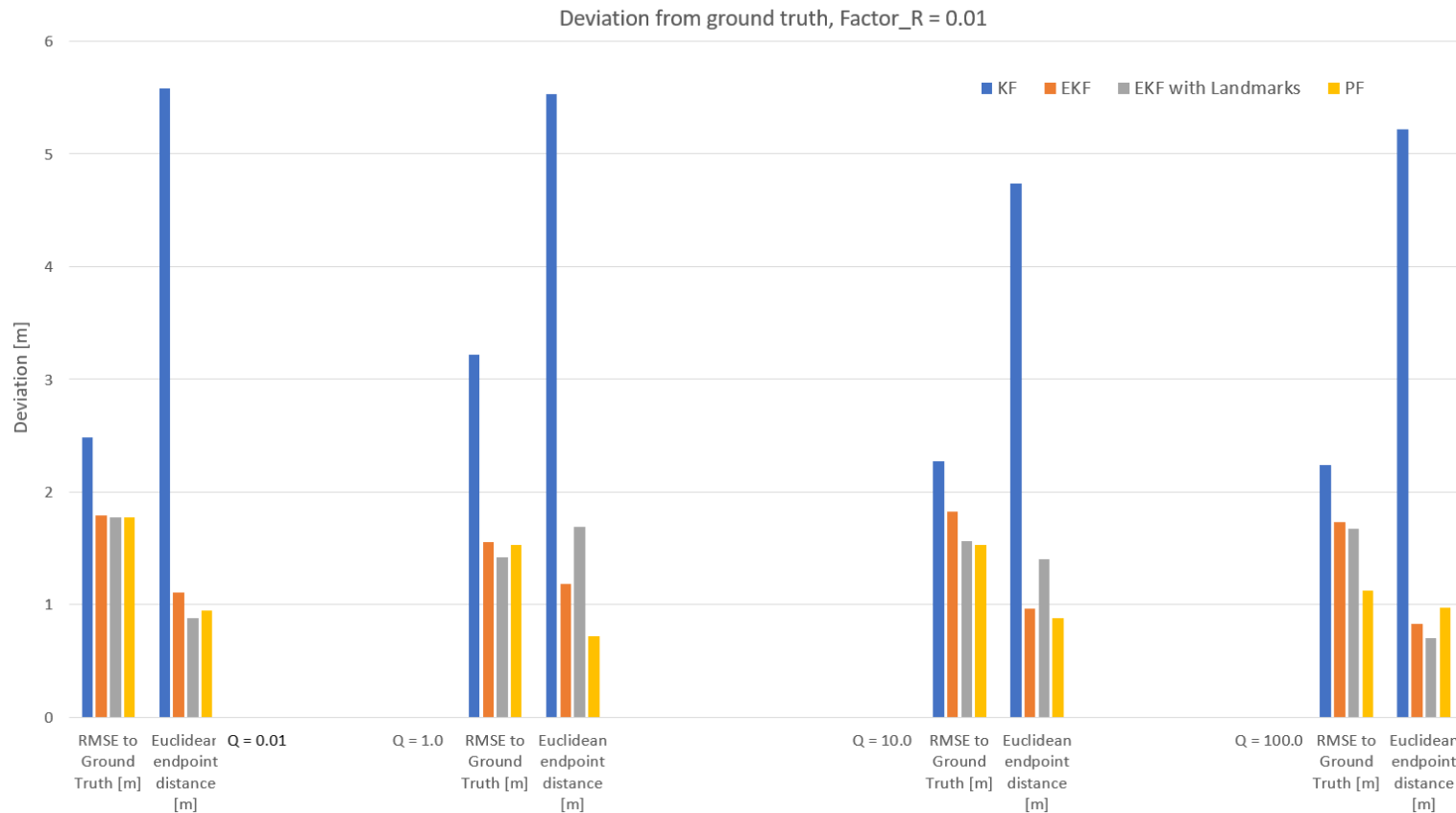


Figure 5: Experimental results of Evaluation Group 1 - R = 0.01.

Results – Evaluation Group 2: Standard Measurement Noise

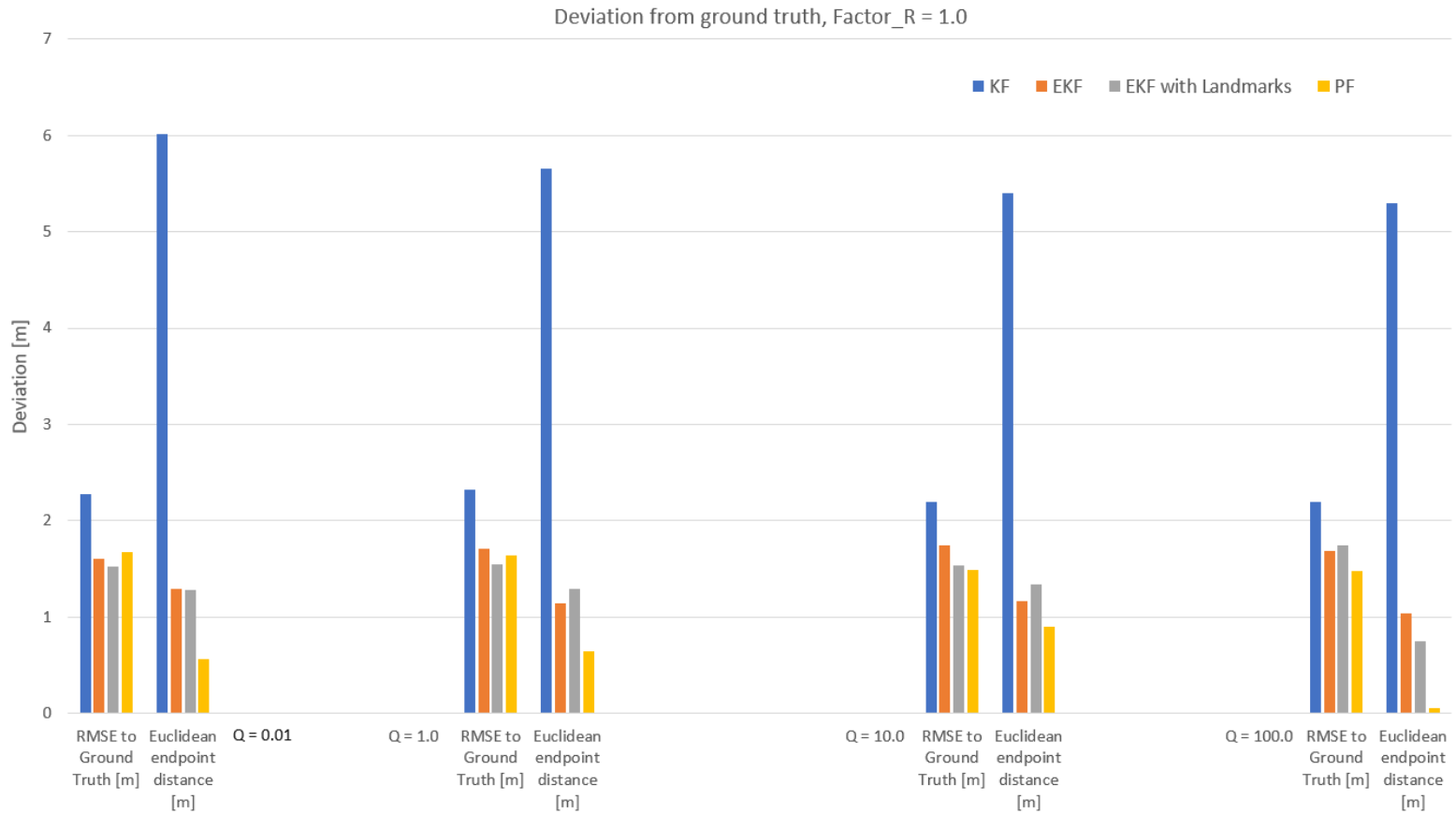


Figure 6: Experimental results of Evaluation Group 2 - R = 1.0.

Results – Evaluation Group 3: Intermediate Measurement Noise

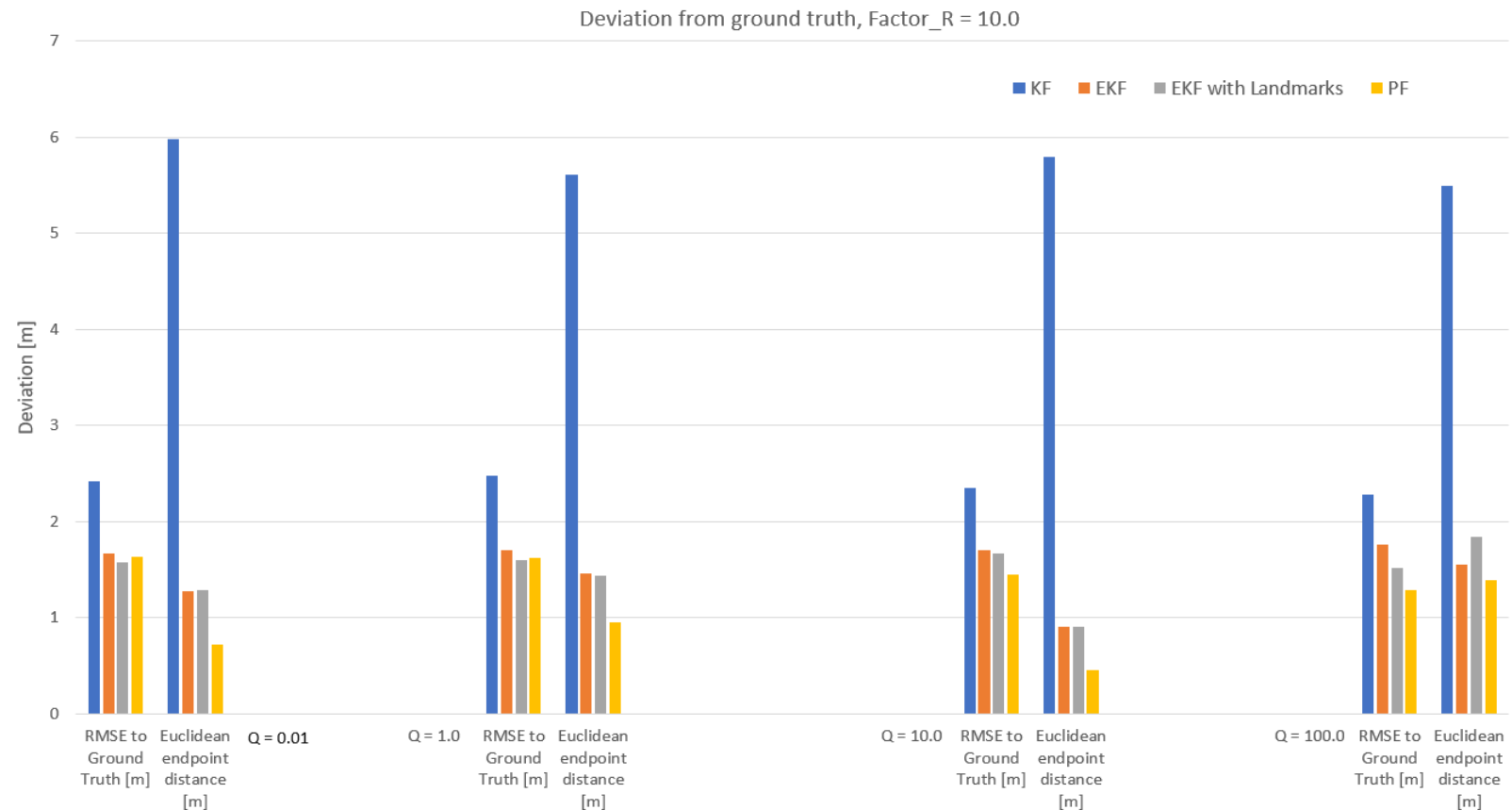


Figure 7: Experimental results of Evaluation Group 3 - R = 10.0.

Results – Evaluation Group 4: Extreme Measurement Noise

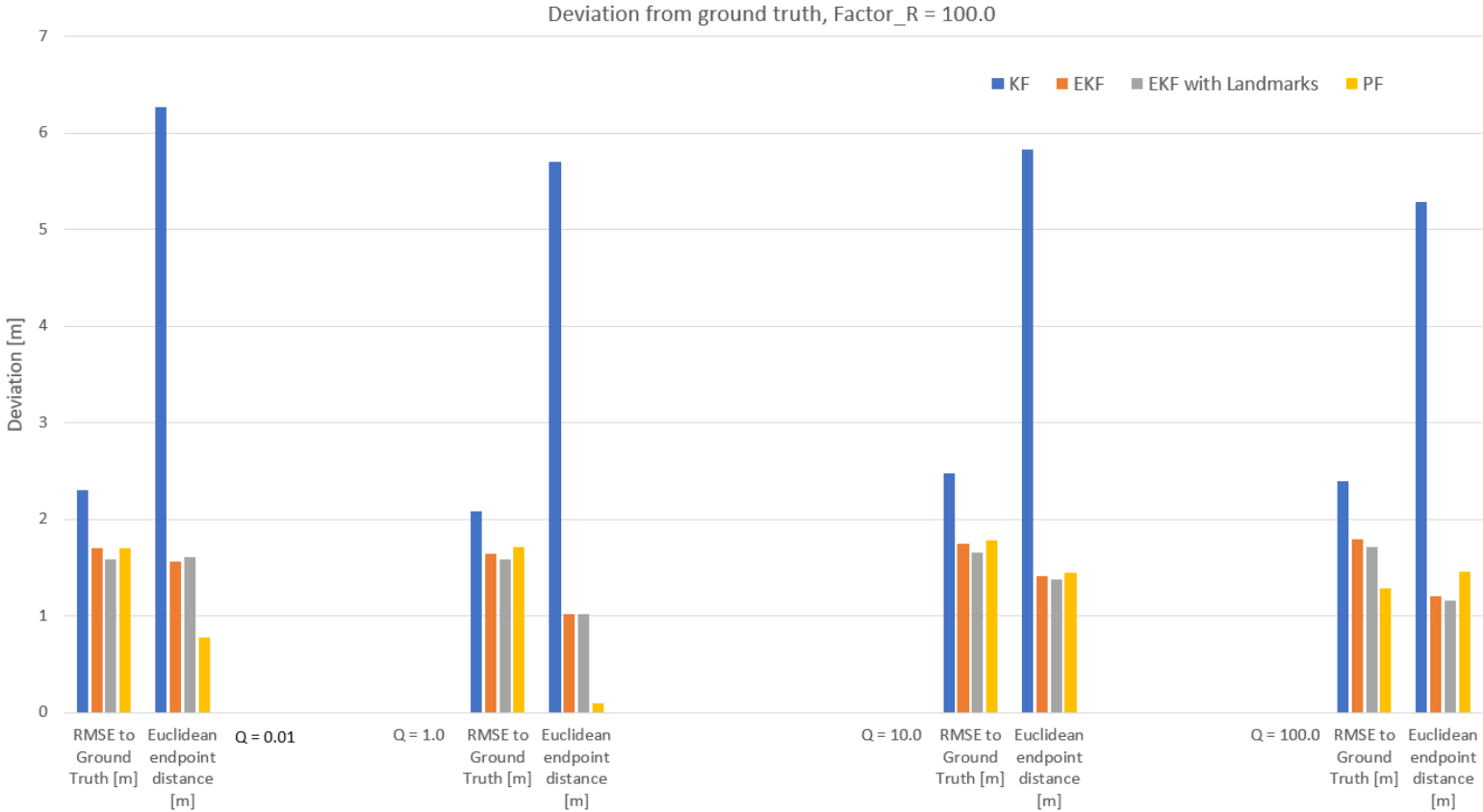


Figure 8: Experimental results of Evaluation Group 4 - R = 100.0.

Discussion

- *KF*
 - Structural bottleneck due to linear motion model
 - Not a parameter tuning issue
- *EKF*
 - Robust under low measurement noise (up to $R = 1.0$)
 - Extreme measurement noise ($R = 100.0$) zeroes out Kalman gain
 - Sensitive to changes in orientation
- *EKF-LM*
 - Landmark updates successfully drift under low noise
 - Inject high-frequency jitter under extreme process noise ($Q = 100.0$)
- *PF*
 - Achieved the highest tracking resilience
 - Extreme Q values induce destructive particle dispersion during sharp turns
 - Physical plausibility check – preventing impossible tracking failures, “driving through walls”

Discussion

Comparison of the estimators

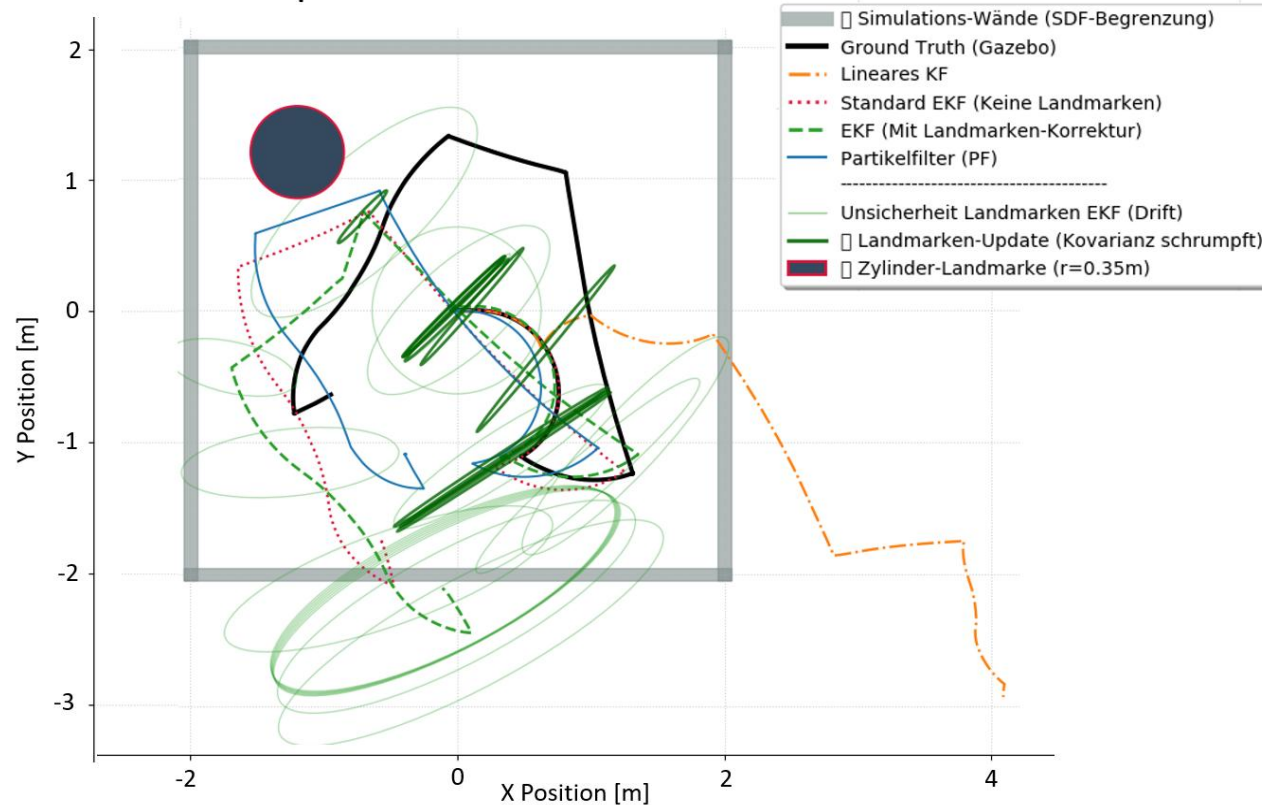


Figure 9: Example run ($Q = 1.0$, $R = 0.01$) where the EKF's estimated the Robot to drive through walls.

Summary and Outlook

- *Conclusion & Structural Trade-offs*

- Evaluated 4 architectures across 16 noise configurations on a TurtleBot simulation
- **KF**: Inherently insufficient due to inability to map angular dynamics
- **EKF**: Quite robust overall, but sensitive to measurement noise ($\rightarrow \theta$)
- **EKF-LM**: Successful under low noise, but injects jitter under extreme Q
- **PF**: Highest resilience against noise, plausibility check

- *Future work & Outlook*

- Physical deployment
- Landmark Refinement with multiple cylinders
- Implementing plausibility checks in all architectures

Thank you for your attention!

Questions?